

Manaswinee Bezbaruah

Curriculum Vitae

Massy, Île-de-France, FR
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EXPERIENCE

- Postdoctoral Researcher** 2025 - Present
POEMS, Unité de Mathématiques Appliquées (UMA)
ENSTA Paris, Palaiseau, France
- Graduate Teaching/Research Assistant** 2019 - 2025
Department of Mathematics,
Texas A&M University, College Station, TX, USA
- Computing Scholar** Summer 2024
MFEM - Finite Element Discretization Library,
Lawrence Livermore National Laboratory, Livermore, CA, USA
- Graduate Research Assistant** Summer 2023
T-5 Division (X-Ray and CT),
Los Alamos National Laboratory, Los Alamos, NM, USA

EDUCATION

- Doctor of Philosophy, Mathematics** Spring 2025
College of Science and Engineering, **Texas A&M University, College Station**
Chair: Dr. Matthias Maier, Co-Chair: Dr. Timothy Davis
- Bachelor of Science (Honors), Mathematics** Spring 2019
College of Science and Engineering, **University of Minnesota, Twin Cities**
Minor: Computer Science

RESEARCH INTERESTS

Numerical methods for PDEs, finite element methods, finite element software, parallel and distributed computing, PDE-constrained optimization, shape optimization, time-harmonic Maxwell's equations.

PUBLICATIONS AND PRE-PRINTS

- M. Bezbaruah, M. Maier, W. Wollner (2024)
Shape optimization of optical microscale inclusions.
SIAM Journal on Scientific Computing, 46(4), B377–B402,
Available at: <https://doi.org/10.1137/23m158262x>
- T. Mattson, M. Bezbaruah, M. Maier, S. McMillan, M. Pelletier, E. Welch, T. Davis (2024)
Indexed Binary Operations in the GraphBLAS.
IEEE HPEC 2024 [Conference Proceeding],
Available at: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10938456&tag=1>

SOFTWARE DEVELOPMENT

Time-harmonic Maxwell's Equations Solver, deal.ii step-81
Texas A&M University, College Station, TX, USA

Shape Optimization of Microscale Inclusions, <https://zenodo.org/records/10459309>
Texas A&M University, College Station, TX, USA

RESEARCH PROJECTS

Numerical Analysis of Maxwell's Equations in Cold Plasma 2025 - Present
Mentors: Dr. Patrick Ciarlet and Dr. Maryna Kachanovska
ENSTA Paris, Palaiseau, Île-de-France, France

Parallel Implementation of Finite Element Assembly 2024 - 2025
Mentors: Dr. Matthias Maier and Dr. Timothy A Davis
Texas A&M University, College Station, TX, USA

GPU Implementation of Target-Matrix Optimization Paradigm Algorithms Summer 2024
Mentor: Dr. Ketan Mittal
Lawrence Livermore National Laboratory, Livermore, CA, USA

Eigenvalue Optimization of Microstructures 2023 - 2025
Mentor: Dr. Matthias Maier
Texas A&M University, College Station, TX, USA

Sparse View Tomography: Shape Optimization Summer 2023
Mentors: Dr. Michael McCann and Dr. Marc Klasky
Los Alamos National Laboratory, Los Alamos, NM, USA

Shape Optimization of Microstructures 2022 - 2023
Mentors: Dr. Matthias Maier and Dr. Winnifred Wollner
Texas A&M University, College Station, TX, USA